

Nicolas De Angelis

Curriculum Vitae

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Born 05 November 1997



Currently employed as a postdoctoral researcher at INAF-IAPS (Italian National Institute of Astrophysics - Institute of Space Astrophysics and Planetology) in Rome (IT), my research interests include high-energy (X/gamma-ray) polarimetry astrophysics with space-based missions applied to transient sources such as Gamma-Ray Bursts (GRBs) or Solar Flares (SFLs) both on an instrumentation and data-analysis point of view. In parallel, I am also interested in spreading science to the world through teaching and outreach.

Research Experience

Apr 2026 – **Fixed-term Researcher**, *Istituto di Astrofisica e Planetologia Spaziali (IAPS)*,
Mar 2028 *National Institute for Astrophysics (INAF)*, Rome, Italy.

Feb 2024 – **Post-doctoral researcher**, *Istituto di Astrofisica e Planetologia Spaziali (IAPS)*,
Mar 2026 *National Institute for Astrophysics (INAF)*, Rome, Italy.

Development of experiments for X and Gamma Ray Astronomy from space for the CUSP, SWIPE, PRIN MIUR 2020 Hype-X, IXPE, eXTP and HERMES projects

Optical and High-Energy Detector characterization and testing, Electronics testing and debugging, Instrument design and development for hard X-ray polarimetry

Outreach activities: IAPS Open Days 2024, European Nights of Research Rome 2024, CNR-INAf Open Day for Schools 2025, IAPS Open Days 2025, European Nights of Research Rome 2025

Sept 2019 – **PhD in high energy astrophysics**, *Département de Physique Nucléaire et Corpusculaire (DPNC)*, *University of Geneva*, Geneva, Switzerland.
Dec 2023

PhD thesis: Development of the Next Generation Space-based Compton Polarimeter and Energy Resolved Polarization Analysis of Gamma-Ray Bursts Prompt Emission – <https://doi.org/10.13097/archive-ouverte/unige:173869>

Development of the next generation Compton polarimeter POLAR-2: SiPM characterization, Optical characterization and Geant4 simulation, data acquisition with ASIC/FPGA-based electronics, Space qualification campaigns (irradiation, vibration and shock, thermal vacuum cycling), Mechanical assembly/design, Gamma-ray detector calibration

Gamma-Ray Burst polarization data analysis with the POLAR data

Involved in the development of other future space-based experiments: eXTP, LEAP

Research work published in peer-reviewed journals and presented in international conferences/workshops

Doctoral lectures: Gas detectors, Semi-conductor detectors, Bayesian statistics, Bayesian methods applied to astrophysics, High energy astrophysics, Advanced scientific computing

Outreach activities: Nuit de la Science 2018, Nuit de la Science 2022, Young Researchers Day 2023

- Sept 2019 – **Teaching Assistant**, *Section de Physique, University of Geneva*, Geneva, Switzerland.
June 2023
Teaching experimental laboratory of physics (classical mechanics, thermodynamics, optics, electronics, nuclear physics) to biology students - 8 hours per week (2019-2020)
Teaching general physics (classical mechanics, thermodynamics, electrodynamics, hydrodynamics, elasticity) to biology, geology, pharmacy, and IT students - 4 hours per week (2020-2023)
- Apr 2018 – **Master Thesis**, *Département de Physique Nucléaire et Corpusculaire (DPNC)*,
July 2019 *University of Geneva*, Geneva, Switzerland.
Studies of readout electronics and optical elements for a gamma-ray telescope (CTA SST-1M) – <https://cds.cern.ch/record/2683289>
Work presented at the CTA Consortium meeting of Lugano in June 2019
- Sept–Dec 2018 **Research and Teaching Assistant**, *Département de Physique de la Matière Quantique (DQMP)*, *University of Geneva*, Geneva, Switzerland.
Teaching experimental laboratory of physics to biology students - 4 hours per week
- Summer 2017 **Internship at CERN in the group of Prof. Iacobucci (ATLAS collaboration)**, *European Organization for Nuclear Research*, Geneva, Switzerland.
Finite element simulation of hybrid HV-CMOS cross coupling

Education and Training

- 2019–2023 **PhD degree in Physics**, *Département de Physique Nucléaire et Corpusculaire (DPNC)*, *University of Geneva*, Geneva, Switzerland.
Lectures: Gas Detectors; Semiconductor Detectors; Bayesian Statistics; Bayesian Methods Applied to Astrophysics; High-Energy Astrophysics; Advanced Scientific Computing
- 2017–2019 **Master's degree in Nuclear and Particle Physics**, *Département de Physique Nucléaire et Corpusculaire (DPNC)*, *University of Geneva*, Geneva, Switzerland.
Advanced Particle Physics I (QED - Quantum Electrodynamics); Advanced Particle Physics II (QCD - Quantum Chromodynamics, Weak Interactions, Symmetry Breaking and Higgs Field); General Relativity; Quantum Field Theory; Observational Cosmology; Superconductivity and Applications; Neutrino Physics and LHC physics; Detectors and Accelerators; Experimental Laboratory IV; Seminars of the DPNC (Nuclear and Corpuscular Physics Department).
Master Thesis Research Project; Teaching assistant; Outreach activities
- 2014–2017 **Bachelor's degree in Physics**, *University of Geneva*, Geneva, Switzerland.
Electrodynamics I; Classical Mechanics; Mathematical Methods for Physicists I; Mathematical Analysis; Linear Algebra; Introduction to Computer Science and Programming; Experimental Laboratory I.
Real Analysis; Complex and Fourier Analysis; Thermodynamics; Hydrodynamics; Elasticity Analytical Mechanics; Quantum Mechanics; Electrodynamics II; Mathematical Methods for Physicists II; Experimental Laboratory II.
Introduction to astronomy & astrophysics; Advanced Quantum Mechanics; Statistical Mechanics; Introduction to Particle and Nuclear Physics; Solide State Physics; Experimental Laboratory III; Seminars; Planetary science; High-Energy Astrophysics; Physical bases of astrophysics

2011–2014 **French Baccalaureat in Science (A-Level equivalent) with Honours, major in engineering sciences and mathematics**, *Lycée Charles Poncet*, Cluses, France.

Relevant Skills

Software	$\LaTeX$; Matlab; Mathematica; Cadence PSpice; LabView; COMSOL Multiphysics; SolidWorks; Catia; mecaflux Heliciel; Igor Pro; Maple; Adobe Illustrator/Lightroom; Gimp; Microsoft Office Suite
Programming language	Python; C; C++; ROOT; Geant4; Bash; basic knowledge of R, C#, and IDL
Experimental	Work in Clean Room; Radioactive Sources (alpha, beta, gamma, neutron); Synchrotron Facilities; X-Ray Tubes; Radiation Protection; Instrument Design and Development, Assembly, and Calibration; Space Qualification: Irradiation, Thermal Vacuum Cycling, Vibration and Shock Test, UV irradiation, Outgassing; Optical Characterization
Simulation	Geant4 Radiation Simulation; Geant4 Optical Simulation; Electronics PSpice Simulation; Multi-Physics Simulation; Monte Carlo Simulation
Data Analysis	Forward Folding; threeML; Multi-Likelihood Analysis; Spectro-Polarimetric analysis; Bayesian Analysis; Signal Processing; HEASoft and FTOOLS
Teaching, Outreach, Supervision	Teaching General Physics Bachelor-level Courses; Teaching Astrophysics/Instrumentation Master-level Courses; Teaching Bachelor Physics Laboratory Courses; Presenting Research at Public Events; Development of Animations and Figures for Outreach; Supervision of Engineering internship students; Supervision of Physics Summer Student from CERN
Paper Reviewing	Paper Referee for Peer-Reviewed Journals

Languages

French	Mother Tongue
English	Fluent
Italian	Fluent
German	Basic
Russian	Basic

Basic words and phrases only

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Nicolas De Angelis,
27th February 2026